Instructions: Instructions: Read each question and examine the included diagrams carefully. Select the best answer from the six available options.

Refer to the notes here to help answer the questions: Vector Kinematics and Dynamics
Offline Paper HW: U1HW8 - W11 - Vector Kinematics and Dynamics 1_questions.pdf

1. What is the primary focus of **dynamics**?

- The study of why objects move, examining how forces cause objects to accelerate, slow down, or turn.
- The mathematical description of motion without considering the underlying forces.
- The measurement of an object's total mass without spatial coordinates.
- The study of stationary fluids and hydrostatic pressure.
- The calculation of electromagnetic waves in a vacuum.
- The study of chemical reactions in dissolving solutes.

Answer:

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2. How does **vector kinematics** differ fundamentally from dynamics?

- Vector kinematics measures the energy consumed by an engine.
- Vector kinematics describes how objects move using directional measurements like displacement and velocity without focusing on the forces causing motion.
- Vector kinematics strictly measures scalar quantities such as mass and temperature.
- Vector kinematics focuses exclusively on why gravity acts on celestial bodies.
- Vector kinematics ignores spatial direction and only considers numerical magnitude.
- Vector kinematics only applies to objects at complete rest.

Answer:

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3. In the International System of Units (SI), what is the standard scientific unit used to measure **force**?

- Joule (J)
- Kilogram (kg)
- Newton (N)
- Meter per second (m/s)
- Watt (W)
- Pascal (Pa)

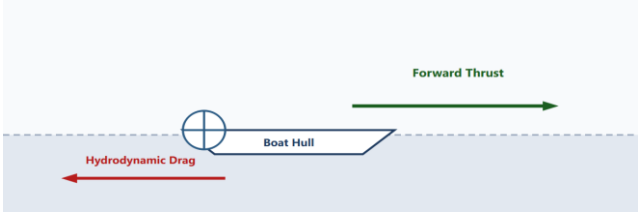
Answer:

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4. Examine the propulsion diagram below: According to the text, how is the **Forward Thrust** generated?



- The rotating paddle blades push backward against the water, and the water exerts an equal forward reaction force on the boat.
- The engine pulls the surrounding air forward into the hull.
- Hydrodynamic drag pushes the boat forward as speed increases.
- The water pulls the boat forward through capillary suction.
- Gravity pulls the boat horizontally across the surface of the water.
- Buoyant forces eliminate the need for any propulsive thrust.

Answer:

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5. In fluid mechanics and boat propulsion, in what direction does **hydrodynamic drag** act relative to the hull's motion?

- Perpendicular upward toward the sky.
- Directly opposite to the direction of the hull's forward motion.
- In the exact same direction as the forward thrust.
- Perpendicular downward toward the seabed.
- At a 45-degree angle pointing northeast at all times.
- Hydrodynamic drag has magnitude only and no physical direction.

Answer:

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6. What does the term **magnitude** strictly denote in physical measurements?

- The compass bearing or spatial direction of a vector.
- The rate at which an object changes its heading.
- The numerical size, amount, or physical quantity expressed by a number and unit, independent of direction.
- The total time elapsed during a physical event.
- The friction coefficient between two touching surfaces.
- The ratio of mass to volume in a substance.

Answer:

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7. In the recorded laboratory measurement **'18 Newtons upward'**, which component represents the **magnitude**?

- Upward
- 18 Newtons (or 18 N)
- Newtons upward
- Zero
- Vertical plane
- The upward acceleration

Answer:

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8. Inspect the vector diagrams below:
In scientific illustrations, how is the **magnitude** of a vector visually indicated?



- By the color palette used for the arrow.
- By the physical length of the arrow.
- By the angle the arrow makes with the horizontal axis.
- By the dotted line pattern.
- By whether the arrow points left or right.
- By the font style of the text label.

Answer:

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9. Referring to Condition A (2 N) and Condition B (10 N) shown in the vector diagram of Question 8, how many times greater is the magnitude of the force in Condition B compared to Condition A?

- 2 times greater
- 3 times greater
- 5 times greater
- 8 times greater
- 10 times greater
- 20 times greater

Answer:

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10. Which statement precisely characterizes a **scalar quantity**?

- It has magnitude only and is completely described by a number and a unit without direction.
- It must always have both a magnitude and a spatial compass direction.
- It can only describe moving objects in three dimensions.
- It is always represented with an arrow pointing toward the origin.
- It changes value whenever an object rotates in space.
- It is quantified solely in units of Newtons.

Answer:

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